Reconstructing Curves Using the Crust

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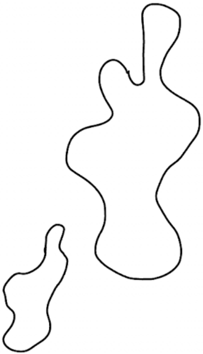
March 21, 2026

Polygonal Reconstruction of a Curve

- Points are taken from a smooth curve, known as **sampling**
- Segments are drawn to connect all adjacent samples, and connect no other points
- Under the correct sampling conditions, algorithms can guarantee a successful reconstruction



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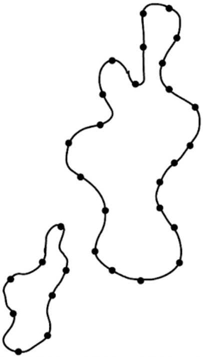

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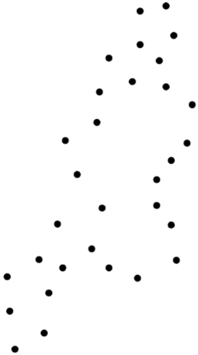

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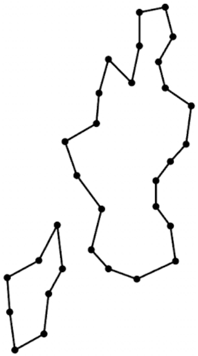

Geometric Tools

Triangulations

- A triangulation creates a mesh that connects all points in a set using only triangles
- Any set of points can be triangulated, usually in many different ways
- Some triangulations are more useful than others

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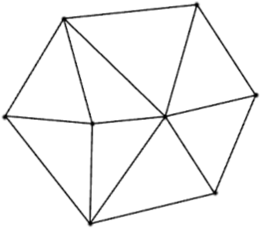
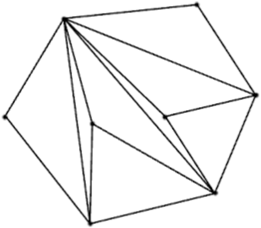
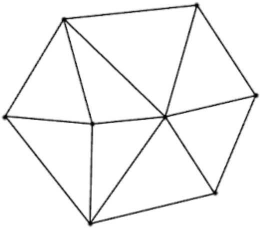
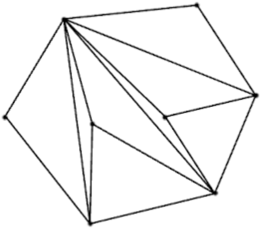




Delaunay Triangulations

- Delaunay triangulations obey the **empty circle property**
- Creates a balanced triangulation, reducing the number of thin triangles
- Always exists and is unique for a given set of points





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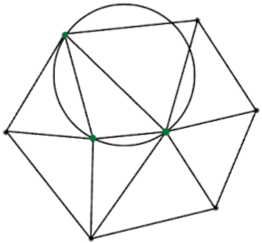
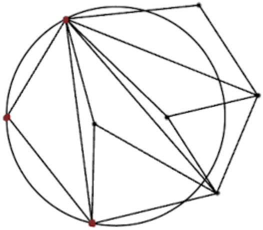
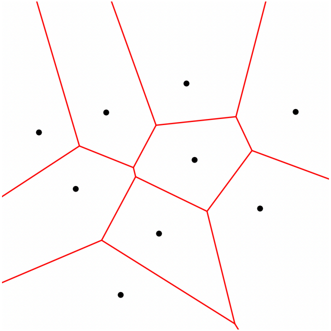



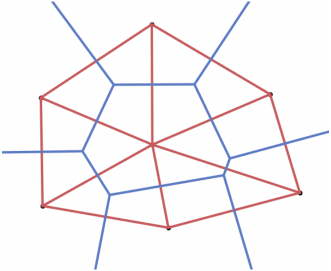
Voronoi Diagrams

- The original set of points are now called **sites**
- Divide the plane into **regions of proximity**
- Consists of Voronoi edges and Voronoi vertices
- Useful to quickly find the nearest site for any given point in the plane



The Delaunay-Voronoi Relationship

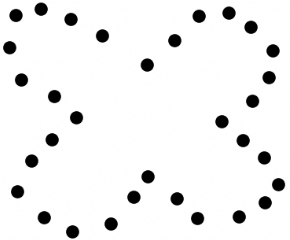
- Voronoi vertices are equidistant to at least 3 sites
- A circle centered at a Voronoi vertex connecting these sites contains no other sites
- Drawing lines between these sites yields a triangle that satisfies the empty circle property
- The Voronoi diagram and the Delaunay Triangulation are **dual graphs** of one another



The Crust

The Crust

1. Start with a set of points S and compute the Voronoi Vertices V.
2. Let $S' = S \cup V$ and let E be the set of edges of the Delaunay triangulation of S'
3. All edges in E that connect two points of S make up the crust



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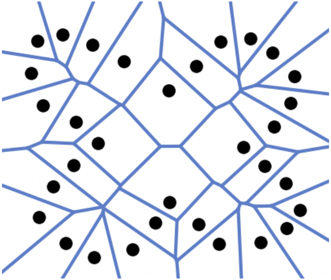

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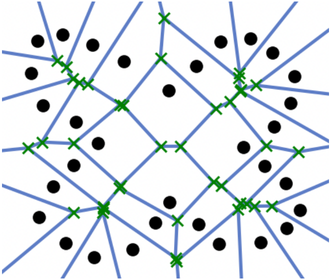

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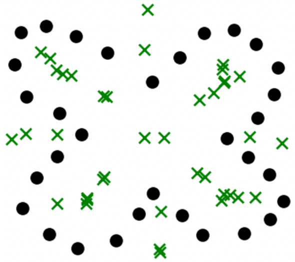

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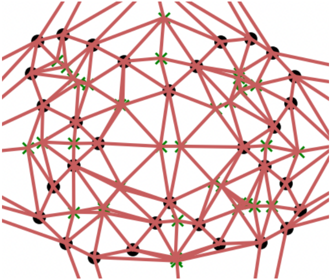

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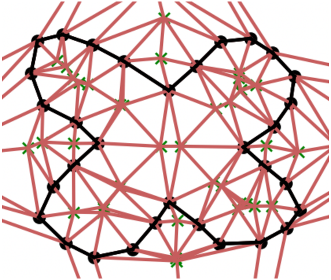

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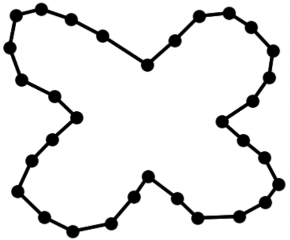
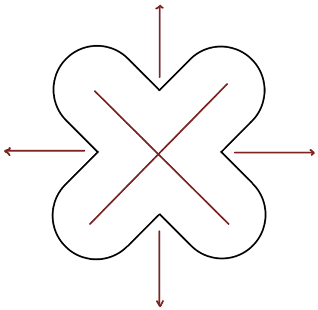


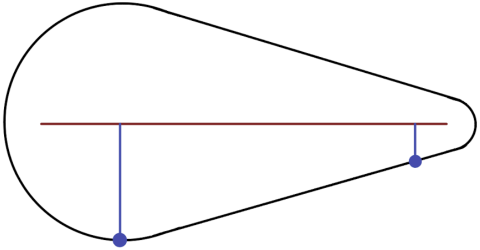
The Medial Axis

- The medial axis is similar to a Voronoi diagram but for an infinite set of points
- The medial axis is the set of all points in the plane that are nearest to more than one point of the smooth curve.
- This definition allows the medial axis to serve as a sort of midline throughout a shape



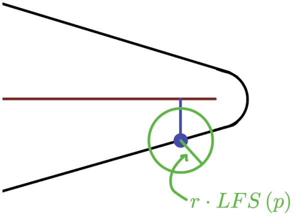
Local Feature Size

- At a point p on a smooth curve, we can define the local feature size, $LFS(p)$ as the distance to the closest point of the medial axis
- As the name suggests, the local feature size gives an idea of how big the features of a shape are at any given point
- When a curve is more detailed, LFS is small, and when a curve is very flat it is large



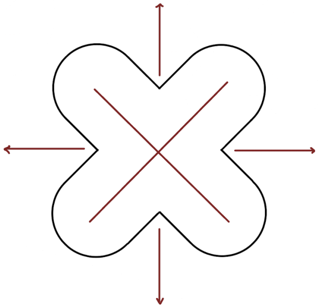
Sampling

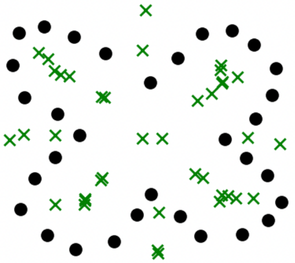
- We say that a curve is r-sampled if the distance between any pair of adjacent samples is less than or equal to $r \cdot LFS(p)$
- This means that a more detailed section of a curve has samples very close together, while less detailed sections can be far apart
- An r-sampled curve with $r < .252$ is guaranteed to be successfully reconstructed by the crust



Why Does the Crust Work?

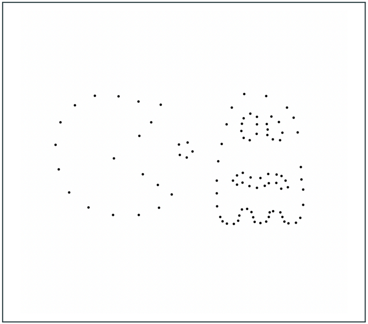
- The Voronoi vertices approximate the medial axis
- A circle connecting two adjacent points in the crust must not contain a Voronoi vertex by the empty circle property
- This means that the crust can't "cut across the middle" of the shape

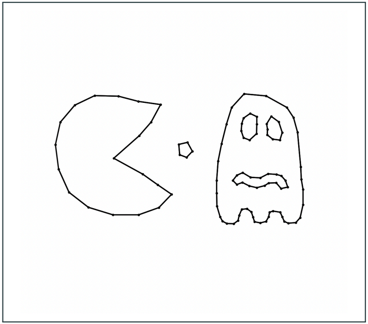




Conclusion

- The crust uses foundational tools to efficiently reconstruct a curve
- If you can approximate "how detailed" a shape is going to be, you can sample accordingly





Thank you!

Questions?